

MSTAR PRACTICE WORKSHEET

Lesson 5-1: Determine the Area of Parallelograms and Rhombuses

Grade 6 Mathematics · Standard 6.GR.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is the area of a parallelogram with base 10 cm and height 7 cm?

- ☐ A 17 sq cm
- ☐ B 34 sq cm
- ☒ C 70 sq cm
- ☐ D 70 cm

 $A = b \times h = 10 \times 7 = 70$ square centimeters.

Question 2 SELECTED RESPONSE — CORRECT: A

A parallelogram has an area of 54 sq ft and a base of 9 ft. What is the height?

- ☒ A 6 ft
- ☐ B 27 ft
- ☐ C 45 ft
- ☐ D 63 ft

 $A = b \times h \rightarrow 54 = 9 \times h \rightarrow h = 54 \div 9 = 6$ ft.

Question 3 **SELECTED RESPONSE — CORRECT: C**

What is the area of a parallelogram with base 9 cm and height 5 cm?

- ☐ A 14 sq cm
- ☐ B 22.5 sq cm
- ☒ C 45 sq cm
- ☐ D 90 sq cm

$A = b \times h = 9 \times 5 = 45$ square centimeters.

Question 4 **SELECTED RESPONSE — CORRECT: C**

Which measurement is the height of a parallelogram?

- ☐ A The length of the slanted side
- ☐ B The perimeter divided by 4
- ☒ C The perpendicular distance between the base and the opposite side
- ☐ D The longest side

The height is always the perpendicular (straight up-and-down) distance between the base and the opposite side, not the slanted side.

Question 5 **SELECTED RESPONSE — CORRECT: D**

Cut the slanted end off a parallelogram and slide it to the other side. What shape forms, and why does that show $A = \text{base} \times \text{height}$?

- ☐ A A bigger parallelogram, so the area gets larger
- ☐ B A triangle, so the area is cut in half
- ☐ C A square, so all four sides must be equal
- ☒ D A rectangle with the same base and height, so its area equals base x height

Sliding the cut triangle turns the parallelogram into a rectangle that keeps the same base and height. A rectangle's area is base x height, so the parallelogram's area is base x height too.

Question 6

SELECTED RESPONSE — CORRECT: C

A blueprint parallelogram has a base of 8 ft, a height of 5 ft, and a slanted side of 7 ft. What is its area?

- ☐ A 26 sq ft
- ☐ B 35 sq ft
- ☒ C 40 sq ft
- ☐ D 56 sq ft

Area = base $\times$ height = $8 \times 5 = 40$ sq ft. The 7 ft slanted side is not used; only the perpendicular height belongs in the formula.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: D

A parallelogram-shaped patio has a base of 16 feet and a perpendicular height of 9 feet. What is its area?

- ☐ A 25 square feet
- ☐ B 50 square feet
- ☐ C 72 square feet
- ☒ D 144 square feet

Area = base $\times$ height = $16 \times 9 = 144$ square feet.

- **A:** That adds 16 and 9. Area multiplies the two measurements.
- **B:** That is the perimeter, 2 times 16 plus 9. Area multiplies base by height.
- **C:** That halves the product, which is the TRIANGLE rule. A parallelogram is base times height with no halving.

Part B — correct answer: D

The slanted side of that patio measures 11 feet. Why is 11 feet NOT used in the area calculation?

- ☐ A Because 11 is an odd number.
- ☐ B Because the slanted side is always longer than the height.
- ☐ C Because only the base matters in a parallelogram.
- ☒ D Because the height must be perpendicular to the base, and the slanted side is not.

Area = $b \cdot h$ requires the PERPENDICULAR height. The slanted side is longer than the perpendicular distance between the parallel bases, so using it would overstate the area.

Question 8

SELECT ALL THAT APPLY — CORRECT: A, B, D, E

Select ALL statements that are true about finding the area of a parallelogram:

- ☒ **A** Any side can be chosen as the base, as long as the height is perpendicular to that side.
- ☒ **B** The height is the distance straight across between the two parallel bases.
- ☐ **C** The slanted side can be used as the height.
- ☒ **D** A rectangle is a parallelogram, so $A = b \cdot h$ works for it too.
- ☒ **E** Doubling only the base doubles the area.

A, B, D, and E are true. C is false: the slanted side is longer than the perpendicular height, so using it overstates the area.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A parallelogram has a base of 12 cm. If you double the height, the area doubles too. Explain why this happens using the formula, and give a specific example with numbers.

Model answer: In $A = b \times h$ the base of 12 cm is a fixed factor, so the area is just 12 times whatever the height is. With $h = 5$, $A = 12 \times 5 = 60$ sq cm; doubling to $h = 10$ gives $A = 12 \times 10 = 120$ sq cm, exactly double, because doubling one factor doubles the product.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.